

Almat Akhmetali

almat.akhmetali@nu.edu.kz • <https://almatakhmetali.github.io> • +7 707 280 49 92
Nazarbayev University, Astana, Kazakhstan

Education

- | | |
|-------------------|--|
| 08/2026 – Present | Nazarbayev University – Astana, Kazakhstan
Ph.D. in Physics
Advisors: E. Abdikamalov, J. A. Font <i>GPA: $\sim$4.00.</i> |
| 09/2024 – 06/2026 | Al-Farabi Kazakh National University – Almaty, Kazakhstan
M.S in Physics and Astronomy
Advisor: N. Ussipov <i>GPA: 3.88/4.00.</i> |
| 09/2020 – 06/2024 | Al-Farabi Kazakh National University – Almaty, Kazakhstan
B.S. in Physics and Astronomy (with distinction)
Advisor: N. Ussipov <i>GPA: 3.75/4.00.</i> |

Experience

- | | |
|-------------------|---|
| 01/2026 – Present | Research Assistant , Nazarbayev University
Astana, Kazakhstan
7 months |
| 09/2025 – 07/2026 | Engineer , Institute of Experimental and Theoretical Physics, Farabi University
Almaty, Kazakhstan
11 months |
| 03/2023 – 12/2025 | Research Assistant , Farabi University
Almaty, Kazakhstan
2 years 10 months |

Internships

- | | |
|-------------------|--|
| 02/2026 – 02/2026 | Energetic Cosmos Laboratory ,
Astana, Kazakhstan |
|-------------------|--|

06/2024 – 06/2024 **Northwestern Polytechnical University,**
Xi'an, China

04/2024 – 04/2024 **Fesenkov Astrophysical Insitute,**
Almaty, Kazakhstan

Scientific Interests

Core-collapse supernovae

Gravitational waves. Statistical and machine learning analysis of the signals.

Machine learning. Applications of ML in astrophysics.

Star clusters. Exploring fractality of star clusters.

Data analysis.

Awards

- 2026 Certificate of appreciation for outstanding contribution as an Organizer of the 10th international scientific conference "Modern advances in physics and fundamental physics education". April 29-30, 2026
- 2024 2nd place in International Scientific Conference "Farabi Alemi" April 4-6, 2024.

Selected publications (*h*-index: 7. Citations: 95. Source: Google Scholar)

- 2026 **Parameter Estimation Horizon of Core-Collapse Supernovae with Current and Next-Generation Gravitational-Wave Detectors**
A. Akhmetali, Y.S. Abylkairov, D. Orel, S. Nunes, A. Sakan, A. Zhunuskanov, M. Zaidyn, N. Ussipov, J.A. Font, and E. Abdikamalov.
Physical Review D Accepted. DOI: <https://doi.org/10.1103/xxnn-chs6> (**Q1**)
- 2026 **Toward More Realistic Machine-Learning Inference of the Dense-Matter Equation of State from Supernova Gravitational Waves**
A. Akhmetali, Y.S. Abylkairov, M. Zaidyn, A. Sakan, A. Zhunuskanov, N. Ussipov, J.A. Font, A. Torres-Forné, and E. Abdikamalov.
Physical Review D Accepted. DOI: <https://doi.org/10.1103/h2b3-gdix> (**Q1**)

- 2026 **Evolution of fractality in centrally concentrated young clusters**
A. Akhmetali, A. Assilkhan, M.M. Mac Low, N. Ussipov, M. Zaidyn, E. Abdikamalov,
A. Sills, X. Pang and B. Shukirgaliyev.
Astronomy & Astrophysics. DOI: <https://doi.org/10.1051/0004-6361/202659843> (Q1)
- 2026 **Fractality of open clusters in singles, pairs, and groups**
A. Akhmetali.
Journal of Astrophysics and Astronomy. DOI: <https://doi.org/10.1007/s12036-025-10122-3> (Q2)

Full list of publications can be found here: [Google Scholar](#) [ADS](#) [Scopus](#) [Inspire](#)

Conferences, Talks and Visits

- 18-22 April 2026 Centrally-concentrated star formation: star cluster fractality
14th Silk Road Meeting and 2nd Meeting of the Silk Road Astronomy Research & Education Alliance. **Contributed talk**
- 11-15 May 2026 Parameter Estimation Horizon of Core-Collapse Supernovae with Current and Next-Generation Gravitational-Wave Detectors
The Universe at All Scales. **Poster presentation**
- 19 February 2026 Probing nuclear equations of state of neutron stars with gravitational waves
Energetic Cosmos Laboratory, Nazarbayev University. **Seminar**
- 15 July 2025 Fractal Nature of Star Clusters
Energetic Cosmos Laboratory, Nazarbayev University. **Seminar**

Skills

Python, Machine learning, Data analysis, Statistics.